

Data quality checks

Row Counts

1. olist_customers_dataset - 99441
2. olist_geolocation_dataset – 1000163
3. olist_order_items_dataset – 112650
4. olist_order_payments_dataset – 103886
5. olist_order_reviews_dataset – 99224
6. olist_orders_dataset – 99441
7. olist_products_dataset - 32951
8. olist_sellers_dataset – 3095

Missing values

1. olist_customers_dataset - NOT FOUND
2. olist_geolocation_dataset – NOT FOUND
3. olist_order_items_dataset – NOT FOUND
4. olist_order_payments_dataset – NOT FOUND
5. olist_order_reviews_dataset – FOUND
 - review_comment_title – 87656
 - review_comment_message – 58247.
6. olist_orders_dataset – FOUND
 - order_approved_at – 160
 - order_delivered_carrier_date – 1783
 - order_delivered_customer_date – 2965
7. olist_products_dataset - FOUND
 - product_category_name - 610
 - product_name_lenght – 610.
 - product_description_lenght - 610
 - product_photos_qty - 610
 - product_weight_g - 2

- product_length_cm - 2
- product_height_cm - 2
- product_width_cm - 2
- 8. olist_sellers_dataset – NOT FOUND

Duplicate values

1. olist_customers_dataset - NOT FOUND
2. olist_geolocation_dataset – FOUND - 261831
3. olist_order_items_dataset – NOT FOUND
4. olist_order_payments_dataset – NOT FOUND
5. olist_order_reviews_dataset – NOT FOUND
6. olist_orders_dataset – NOT FOUND
7. olist_products_dataset - NOT FOUND
8. olist_sellers_dataset – NOT FOUND

DATA QUALITY & CLEANING ANALYSIS REPORT

Executive Summary

- Comprehensive data quality assessment conducted across the dataset
- Duplicate records identified and removed to improve reliability
- Missing value analysis performed to ensure completeness
- Data consistency and integrity validated before analysis
- Dataset transformed into a clean, analysis-ready format
- Focus: Improving data accuracy, reliability, and usability

High-quality data is the foundation of effective analytics and business decision-making. This project focused on assessing the overall quality of the dataset,

identifying data issues, performing cleaning operations, and preparing the data for accurate downstream analysis.

The initial phase involved a comprehensive data quality assessment to evaluate completeness, consistency, uniqueness, and validity across all available variables. This process helped identify potential issues that could impact analytical accuracy and business insights.

Several important quality checks were performed, including:

- Missing value detection and validation.
- Duplicate record identification and removal.
- Consistency checks across categorical variables.
- Statistical validation of numerical attributes.
- Column standardization and formatting improvements.
- Dataset structure optimization for analysis readiness.

The analysis confirmed strong overall data completeness, with minimal issues related to missing information. However, duplicate records were identified and removed to eliminate redundancy and improve dataset integrity. This cleaning process ensured that future analytical results would accurately represent real-world observations without distortion caused by repeated entries.

Additional transformations were applied to improve usability and readability. Column naming conventions were standardized, data formats were reviewed, and variables were organized to support efficient exploratory and business analysis workflows.

The cleaning process produced a refined dataset that was significantly more reliable and suitable for advanced analytics. By addressing quality issues at an early stage, the project established a trustworthy foundation for subsequent business analysis and reporting activities.

Key Achievements

- Improved dataset reliability through systematic quality assessment.
- Eliminated duplicate records that could bias analytical results.
- Enhanced data consistency and structure for easier analysis.
- Increased confidence in business insights generated from the dataset.

- Created a clean, analysis-ready dataset suitable for reporting and visualization.
- Demonstrated a structured approach to data preparation and governance.

Business Impact

Reliable data directly influences the quality of business decisions. By improving data accuracy, consistency, and integrity, this project reduced the risk of misleading insights and established a stronger foundation for future analytics initiatives. The resulting dataset enables more trustworthy reporting, better forecasting, and improved decision-making across business functions.

Overall, the project successfully transformed raw data into a high-quality analytical asset, highlighting the critical role of data quality management in the analytics lifecycle and reinforcing the importance of proper data preparation before conducting business analysis.